Singapore & Asian Schools Math Olympiad

Secondary 3 (Grade 9) – GEP Practice

2019 & 2020

Contest Problems with Full Solutions

Authors:

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About the Authors:

Henry Ong is the Founder of SASMO and President of Singapore International Mastery Contest Centre (SIMCC). He conducts Math Olympiad classes, not only in Singapore but also around the 20 countries that have joined SASMO. He now travels round the world giving out awards and scholarships to students and teachers. He is now promoting new contests, International Scholastic Trust Teachers' Institute (ISTTI) and International Junior Honor Society (IJHS).

In February 2020, Henry was appointed Co-Chair of the Council Administration of Cambodian Mathematical Society (CA_CMS). Henry will share and use his great wealth of experience, expertise, and international networks to contribute effectively to: (1) promoting STEM: Science, Technology, Engineering, and Mathematics; (2) organizing and undertaking multilateral partnership programs; and (3) strengthening and improving the public-private partnership for development of STEM, research and development, and research and innovation as well as teacher professional development in Cambodia with relevant stakeholders in close collaboration and consultation with relevant leadership and management of Ministry of Education, Youth and Sport, especially Cambodian Mathematical Society (CMS), New Generation Pedagogical Research Center (NGPRC), National Institute of Education(NIE) and Teacher Training Department (TTD).

Merlan Nagidulin is the Academic Director for Math and Science Olympiad and General Manager of SIMCC. Merlan graduated with a Bachelor of Science, Honours in Mathematical Sciences from Nanyang Technological University. Merlan won 2 gold medals in Kazakhstan National Math Olympiad and was awarded a bronze medal in the most prestigious International Math Olympiad (IMO Slovenia 2006).

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International Scholastic Trust President's Message

Dear SIMCC contestants, parents, teachers, and partners,

I hope all of you are safe and healthy. We have all gone through so much within the past year, and we should rejoice - we are all alive and well. Despite all the groom and predictions of the worst recession to hit us very soon, we at SIMCC, are expanding globally – adding another 14 country partners to a total of 35 countries and territories within the last 10 months, and setting up sales offices in Cambodia and India, in addition to taking over the ICAS competitions and staff from the University of New South Wales Global Office based in Hong Kong and Macau.

Two years, Sophie Koh, our Senior Director of IT, Events and Operations left Singapore Polytechnic as a Senior Lecturer of IT to join us. She has almost digitally transformed most of our operations and contest systems. This will provide valuable big data in formative and summative assessments for teachers to measure their students and inform their teaching. Parents and students will also receive Detailed Performance Reports (DPR) in our Member Development Portal (MDP) to improve their Academic Skills. Teachers and their schools will be able to download Students' Analytical Performance Reports (SAPR). SIMCC together with our partners will be conducting more Webinars to prepare teachers and parents to use our contests and data to arm their wards with Skills and Recognition for SUCCESS.

From February 16 to April 30, 2021, we are offering the 1st Singapore Math Global Assessment for grades 1 to 11 students. We want to open this up to every country, especially since Singapore Math textbooks are used in schools from over 70 countries. We want to share our rich Singapore Math heritage that has propelled Singapore students to the top of PISA rankings since the 1990s. We also have excellent teachers, Learning Management System (LMS), Math textbooks, assessment books, as well as teacher training to help improve math education worldwide. Reach out to us for help, if you want your students to become Mathematics Heroes!

The global winners of SINGA Math Assessments are invited for the 1st SINGA Math Global Finals, in their home countries on September 11, 2021. The prize for the Overall Champion of grades 1 to 10/11 is a S$600 competition package to compete in the STEAM AHEAD International Junior Mathematics Olympiad (IJMO) to be held in Bali, Indonesia from December 3 to 7, 2021. The prize for Overall Runner-up is S$250 and 2nd Runner-up is S$150, offering a total prize award of S$10,000. Winners of Gold (10%) and Silver (15%) are invited to compete in STEAM AHEAD IJMO 2021.

From 2021, we will be offering most of our contests to grade 12 students – SASMO, DrCT, SIMOC, and STEAM AHEAD. This will pave the way for Grades 12 SIMCC students to apply for SIMCC and STS's US$270,000 endowed 2 scholarships for 4-year undergraduate studies at Southern Illinois University (SIU), Carbondale, Illinois, USA in STEM related fields. SIU has also given SIMCC over 30 SIU International Student Tuition Grants to award to top SIMCC students to pursue their university education. SIMCC will setup our own network for SIMCC scholars at SIU. This will help us expand SIMCC contests in North America and generate revenue for us to fund more scholarships.

Best Regards,
Henry Ong, President

Competition and Prizes

SASMO is devoted and dedicated to bringing a love for Mathematics to students. Unlike most Math Olympiad Competitions, SASMO caters not only to students in the top 5% but to the top 40% instead. It aims to arouse students' interest and enthusiasm for mathematical problem solving, develop mathematical intuition, reasoning and logical thinking, as well as creative and critical thinking. In addition, this can help improve the students' math grades because they can apply problem-solving strategies learnt during the training to their daily school mathematics.

History:

Created in 2006, SASMO is one of the largest Math Olympiads in the Asian region. We have expanded the competition to provide an International platform for students from Primary 2 to Secondary 4, with differentiated contest papers for every level. SASMO awards medals and certificates to the top 40% of participants.

Calculators are not permitted

When a problem introduces a more advanced concept, all necessary definitions are included.

Awards:

Each participant receives a Certificate of Participation or an award certificate for winners below. Each of the top 8%, 12% and 20% of all participants receives a Gold, Silver or Bronze medal and certificate respectively. Each student who achieves a Perfect Score of 85 points receives a Perfect Score certificate, Gold medal and $100.

Introduction

For Students Taking the Math Olympiad Challenge

Congratulations. You have embarked on a journey of scholarship. Competitions like SASMO open many doors for you. Firstly, you learn new and interesting approaches to problem-solving and new topics. Next, you will meet talented students from other schools as you attend trainings and competitions. You build your endurance to "puzzle" out challenging problems and build your reputation as a problem solver. Finally, you will be exposed to various international competitions and scholarship opportunities. Here in Singapore, you increase your chances of getting into a top school via Direct School Admissions (DSA) and entry into the Gifted Education Programme as you compete regularly in high level competitions.

This book is written for the participants in the Singapore and Asian Schools Math Olympiads (SASMO). It helps students to prepare well for the contest and develop higher-order thinking. All problems are designed to help students develop the ability to think mathematically, rather than to teach more advanced or unusual topics. The fun is in how you can see patterns and ways of solving each problem in non-technical ways even though you have not learnt the topic yet!

In addition to the contest problems, the reader is provided with a list of familiar mathematical terms, as well as a review of some of the topics that are likely to be tested in the Olympiad. The book also contains some solved examples to provide different problem-solving techniques, and to familiarize the participant with different types of Olympiad questions. It is advised that the reader spends appropriate time studying these questions and solutions, as they will assist in tackling actual Olympiad problems.

How to Use This Book: Practice daily for 15 minutes per hour instead of 4 hours of learning once a month. Your mind needs to absorb each new thought, and constant practice allows frequent review of previously learned concepts and skills. Together, you can remember many new problem-solving approaches. Try to spend 10 or 15 minutes daily doing two or three problems. This approach should help you minimize the time needed to develop the ability to think mathematically.

Whether you solve a problem quickly or if you are confused, it is worth studying the solutions in this book, because often they offer unexpected insights that can help you understand the problem more fully. After you have invested time – trying to solve each problem any way you can, reviewing our solutions is very effective. Many of the problems in this book can be solved in more than one way. There is always a single answer, but there can be many paths to that answer. Once you solve a problem, go back and see if you can solve it by another method. Then check our solutions to see if any of them differ from yours.

Enjoy working on these challenges and you will soon be in a different league from your peers who have not taken any international competition. We look forward to inviting you if you are a bronze, silver, gold or perfect score medallist for further training.

Problem Solving Procedure

You may go through several phases when solving a problem such as trying to understand the problem, working on a specific approach (planning and attempting), getting stuck and trying to get unstuck, critically examining solutions or communicating. The work may involve going back and forth between these different phases of problem solving.

In solving any problem, it helps to have a working procedure. You might want to consider this four-step procedure: Understand, Plan, Try It, and Look Back.

Understand

Before you can solve a problem, you must first understand it. Read and re-read the problem carefully to find all the clues and determine what the question is asking you to find. What is the unknown? What is the data? What is the condition?

Plan

Once you understand the question and the clues, it's time to use your previous experience with similar problems to look for strategies and tools to answer the question. Do you know a related problem? Look at the unknown! And try to think of a familiar problem having the same or a similar unknown?

Try It

After deciding on a plan, you should try it and see what answer you come up with. Can you see clearly that the steps are correct?
But can you also prove that the steps are correct?

Are you feeling stuck?

Many different approaches can be tried to get unstuck. One approach is to try working a simpler version of the problem, and use the solution to the problem to get insights that are useful in solving the original problem. In the next chapter, we will show some common solving approaches.

If you are discouraged after a few failed attempts, read this quote from the famous scientist, Thomas Edison. An assistant asked, "Why are you wasting your time and money? We have had failure after failure, almost a thousand of them. Why do you continue to pursue this impossible task?" Edison said, "We haven't had a thousand failures, we've just discovered a thousand ways to not invent the electric bulb."

Look Back

Once you've tried a problem and found an answer, go back to the problem and see if you've really answered the question. Sometimes it's easy to overlook something. If you missed something check your plan and try the problem again.
Can you check the result?
Can you check the argument?
Can you derive the result differently?
Can you see it at a glance?

Problem Solving Strategies

1. Change the representation

Using a wrong representation may make a problem impossible to solve.
Strategies of changing representation include drawing a picture and acting it out.

DRAW A PICTURE: By drawing a picture, and visualizing the problem information using it, you will have a clearer understanding of the problem and it will help you to come up an approach to solve the problem that you might not be able to see otherwise.

ACT IT OUT: We are better at thinking in terms of concrete objects and situations than in terms of abstract concepts. If we can act out the situation described in a word problem, we are able to understand the problem better and we may be able to come up with a solution. To do this, we need to use real materials that are easily available to us. Examples can be pencils, coins and other objects we have in the classroom.

2. Make an Organized List or a Table

ORGANIZED LIST: Making an organized list allows you to clearly examine data. It can help you in ensuring that you are looking at all the relevant information. It will also allow you to see patterns in the data easily and to come to correct conclusions.

MAKE A TABLE: Making a table allows you to clearly examine data. It can help you in ensuring that you are looking at all the relevant information. It will also allow you to see patterns in the data easily and to come to correct conclusions.

3. Create a Simpler Problem

Sometimes we are not able to solve the problem as it is stated, but we are able to solve a similar problem that is similar in some way. For example, the simpler problem may use simpler numbers. Once we solve one or more simpler problems, we may understand the approach that can be used to solve the problems of similar type and may be able to solve the problem that has been given to us.

4. Use Logical Reasoning

Logical reasoning is useful in mathematics problems in various ways. It can be used to eliminate incorrect choices. It can also sometimes be used to conclude the answer directly.

5. Guess and Check

"Guess and Check" strategy can be used on many problems. If the number of possible answers is small, one can use this strategy to come up with the answer very quickly. In some other cases where the number of possible answers is not small, one may still be able to make intelligent guesses and come up with the answer.

6. Working Backwards

Sometimes, it is easier to start with information at the end of the problem and work backwards to the beginning of the problem than the other way around.

SASMO 2019 Secondary 3 (Grade 9) Contest Questions

Section A (Correct answer – 2 points| No answer – 0 points| Incorrect answer – minus 1 point)

Question 1

Evaluate

$$-\frac{1}{4}\times 2019.625 - 0.625 + 0.75 \times 2019.625 + \frac{1}{2}\times 2019.625$$

A. 2019
B. 2019.625
C. 2020
D. -2019
E. None of the above

Question 2

Let x be a positive integer whose prime factors are 5 and 11 only. If x has exactly 6 different positive integer factors, what is the smallest value of x?

A. 110
B. 605
C. 165
D. 275
E. None of the above

Question 3

Suppose $-11 \leq y + 3 \leq 3$ and $0 < x < 8$. What is the greatest possible value of xy?

A. 112
B. 24
C. 0
D. 98
E. None of the above

Question 4

Which of the following is true?

A. $\sqrt{81} = \pm 9$
B. $(x^{12})^{18} = x^{30}$
C. $5x^{-1} = \frac{1}{5x}$
D. $-\frac{2(-x+1)(x+1)}{32} = \frac{1-x^2}{-2^4}$
E. None of the above

Question 5

Let A, M, N, P, S and W be real numbers such that:

$$M = S^1 \times W^2 \times A^3 \times P^4$$

$$N = P^1 \times A^2 \times W^3 \times S^4$$

Suppose $\times A \times S \times P = (1331)^{\frac{1}{15}}$, what is the value of $M \times N$?

A. 11
B. 21
C. 1331
D. Insufficient data
E. None of the above

Question 6

Which of the following is NOT true about a gradient (or slope)?

A. If a line is horizontal, then its gradient is zero.
B. If a line has a negative gradient and the other line has a positive gradient, then these 2 lines intersect at exactly one point.
C. A line with gradient 1 forms a $45°$ angle together with the x-axis.
D. If m_1 and m_2 are gradients of lines such that $m_1 = -m_2$, then the lines are parallel.
E. None of the above

Question 7

Assuming the list below has a mode, what is the greatest difference between the median and mode?

$$\{19, X, 13, 11, 23, 17\}$$

A. 4
B. 5
C. 6
D. 12
E. None of the above

Question 8

Which of the following fractions is the largest?

A. $\frac{654321}{654322}$
B. $\frac{65432}{65433}$
C. $\frac{6543}{6544}$
D. $\frac{654}{655}$
E. $\frac{65}{66}$

Question 9

A 6-digit positive integer which contains digit 8 is divisible by 9. Given all its digits are different, what is the sum of the digits of the least possible value of this integer?

A. 9
B. 19
C. 27
D. 36
E. None of the above

Question 10

If $A = 10^{2019} - 2019$, how many digits of A are odd numbers?

A. 2015
B. 2016
C. 2017
D. 2018
E. None of the above

Question 11

Gryffindor Secondary School has 170 students. A survey of their hobbies shows that:

- 79 like alchemy, 75 like origami and 34 like stage illusion;
- 32 like both alchemy and origami;
- 24 like both alchemy and stage illusion;
- 13 like all the three hobbies;
- 45 have hobbies which are not listed above.

How many Gryffindor students have hobbies which are both stage illusion and origami but not alchemy?

A. 20
B. 7
C. 46
D. 96
E. None of the above

Question 12

In an event, twelve people got 12 different positive integers. The host of the event asked each of them to give a clue about their numbers.

1st person: "My number is bigger than 1."

2nd person: "My number is bigger than 2."

3rd person: "My number is bigger than 3."

…

12th person: "My number is bigger than 12."

Then the host asked them again, and their responses in random order were as follows:

"My number is less than 1."

"My number is less than 2."

"My number is less than 3."

…

"My number is less than 12."

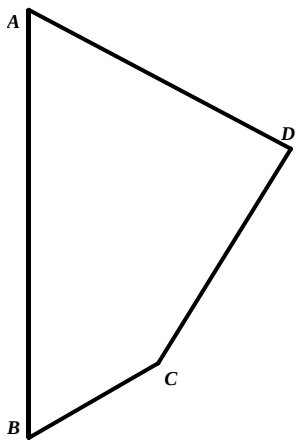
Each person in this event is either a truth-teller (who always tells the truth) or a liar (who always lies). Excluding the host, what is the largest possible number of truth-tellers among these 12 people?

A. 11
B. 10
C. 8
D. 6
E. None of the above

Question 13

In the diagram, $ABCD$ is a quadrilateral with $\angle ABC = \angle BAD = 60°$ and $\angle DAC = \angle DBA$.

Given $AD = 13$ and $BC = 7$, find the length of AB.



A. 20
B. 18
C. $18\sqrt{3}$
D. Insufficient data
E. None of the above

Question 14

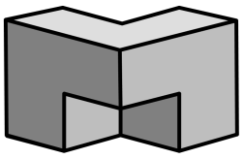
Two calendars are the same if all their dates correspond to the same days. The calendar for the year 2019 is the same for the year ____.

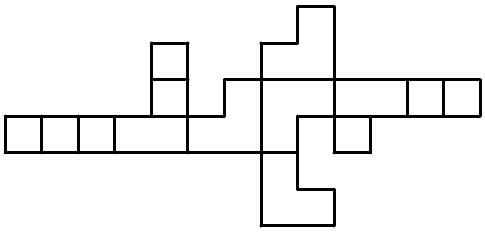
(Years 2020, 2024, 2028 and 2032 are the leap years)

A. 2026
B. 2030
C. 2033
D. 2024
E. 2028

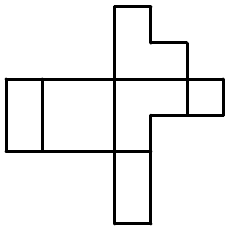
Question 15

Which of the nets below can be folded to form the figure on the right?

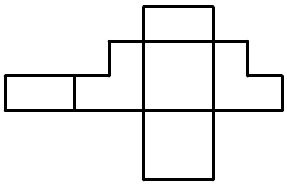




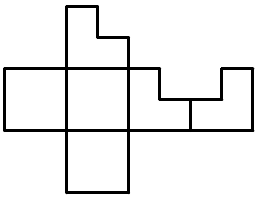
A



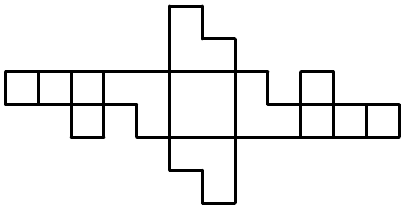
B



C



D



E

Section B (Correct answer – 4 points| Incorrect or No answer – 0 points)

When an answer is a 1-digit number, shade "0" for the tens, hundreds and thousands place.
Example: if the answer is 7, then shade 0007
When an answer is a 2-digit number, shade "0" for the hundreds and thousands place.
Example: if the answer is 23, then shade 0023
When an answer is a 3-digit number, shade "0" for the thousands place.
Example: if the answer is 785, then shade 0785
When an answer is a 4-digit number, shade as it is.
Example: if the answer is 4196, then shade 4196

Question 16

In Mathematics, the product of the first n positive integers is written as $n! = n \times (n-1) \times \ldots 1$. For example, $2! = 2 \times 1$ and $7! = 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1$. What is the remainder when $2018! + 2018$ is divided by 2019?

Question 17

How many 2-digit positive integers are there such that the product of the tens and ones digits is an odd number?

Question 18

The product of 3 consecutive integers is a four-digit number that starts with 4 and ends with 6. What is the sum of these 3 consecutive integers?

Question 19

In the expression below, b is a digit, $\overline{bbbbb}$ is a 5-digit number, and $\overline{bb}$ is 2-digit number. Find the numerical value of the expression in its simplest form.

$$\frac{(b+b)\times\left(\overline{bbbbb}-b\right)-b\times\overline{bb}}{b\times\overline{bb}}$$

Question 20

Find the value of

$$\frac{1-\left(\frac{1}{2019}\right)^2}{1+\left(\frac{1}{2019}\right)^2}+\frac{1-\left(\frac{1}{2018}\right)^2}{1+\left(\frac{1}{2018}\right)^2}+\cdots+\frac{1-\left(\frac{1}{2}\right)^2}{1+\left(\frac{1}{2}\right)^2}+\frac{1-1^2}{1+1^2}+\frac{1-2^2}{1+2^2}+\cdots+\frac{1-2019^2}{1+2019^2}$$

Question 21

The number X is a positive integer with all different digits, and the product of its digits is 504. What is the greatest possible value of $\frac{X}{10}$?

(Round off your answer to the nearest integer.)

Question 22

In the diagram below, $ABCD$ is a rectangle with $AB = 25$ and $AD = 104$. Point E is inside the rectangle such that $DE = 30$ and $CE = 25$. Find the length of AE.



Question 23

In how many ways can the integers 1 through 15 be arranged in a row such that the sum of any two adjacent numbers is a perfect square?

Question 24

In the equation below, x and y are positive integers. What is the greatest possible value of $x + y$?

$$(x^2 + y)(x + y^2) = (x + y)^3$$

Question 25

In the following cryptarithm, all the different letters stand for different digits.

$$\begin{array}{cccccc} & & C & O & O & L \\ + & & M & A & T & H \\ \hline & S & A & S & M & O \\ \hline \end{array}$$

Given that M = 7, find the value of S + A + S + M + O.

END OF PAPER

SASMO 2020 Secondary 3 (Grade 9) Contest Questions

Section A (Correct answer – 2 points| No answer – 0 points| Incorrect answer – minus 1 point)

Question 1

Find the value of the following.

$$2020 \times 2020 - 2020 \times 20 + 2020$$

A. 2020
B. 4042020
C. 4040000
D. 4024020
E. None of the above

Question 2

Which of the following has the largest value?

A. 2020
B. $\frac{2020^{20}}{2020^{18}}$
C. 4080400
D. $\frac{2020^{20}-2020}{2020^{18}-2020}$
E. Options B and C

Question 3

In Mathematics, the product of the first n positive integers is written as $n! = n \times (n-1) \times \ldots \times 1$. For example, $2! = 2 \times 1$ and $7! = 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1$. Find the number of consecutive zeros at the end of $(11! - 10!)$.

A. 1
B. 2
C. 24
D. 26
E. None of the above

Question 4

Find the value of the following expression.

$$\frac{2025 \times (19!)^2 - 5 \times (19!)^2}{100 \times 18! \times 20! + 18! \times 20!}$$

A. 19
B. 2020
C. 2025
D. 2020×19!
E. None of the above

Question 5

Which number below is a factor of $91^4 - 1$?

A. 2019
B. 2020
C. 2021
D. 2022
E. None of the above

Question 6

What is the smallest positive integer that can be expressed as a product of six positive integers such that any two of them are not coprime?

A. 60
B. 64
C. 360
D. 5040
E. None of the above

Question 7

Four friends have accounts in some social networks. Any two of them use at least one common social network, and no social networks are used by more than two of them. At least how many social networks are used by the friends?

A. 4
B. 5
C. 6
D. 8
E. None of the above

Question 8

The value of $f(x) = ax^2 + bx + c$ is 0 when $x = -1$. Given that $f(2) - f(1) = 11$ and $f(3) - f(2) = 15$, what is the value of $a + b + c$?

A. –3
B. –1
C. 1
D. 3
E. None of the above

Question 9

What is the next number in the sequence below?

$$16, 37, 58, 89, 145, 42, \ldots$$

A. 20
B. 31
C. 72
D. 96
E. None of the above

Question 10

There is one counterfeit coin among 2019 identical coins. The genuine coins have the same weight while the counterfeit coin is lighter. Andy wants to determine the counterfeit coin using a balance scale. There are no restrictions to the number of coins placed on the pans each time. What is the least number of weighings needed to find the counterfeit coin?

A. 7
B. 10
C. 11
D. 1009
E. None of the above

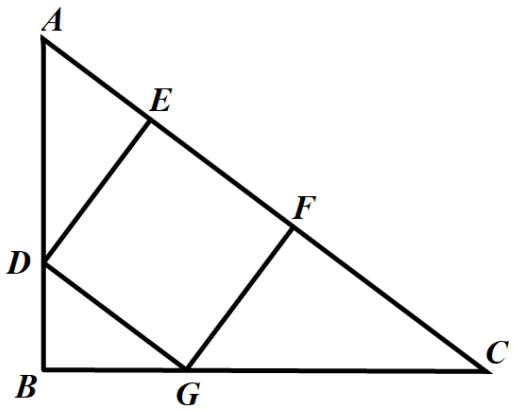
Question 11

How many four-digit numbers have digit 1 but not digit 2 among its digits?

A. 2248
B. 2673
C. 2816
D. 3528
E. None of the above

Question 12

In the diagram, $\angle ABC = 90°$ and $DEFG$ is a square. If $AE = 4$ and $EF = 6$, find the area of quadrilateral $ACGD$.



A. 84
B. 75
C. 63
D. 60
E. None of the above

Question 13

Anthony, Judith, Daniel and Edward were born in 1975, 1976, 1977 and 1978 in the cities of New York, Moscow, Barcelona and Singapore, though not in that order. It is given that

- The person born in Barcelona is one year older than the one born in Singapore.
- Edward was born one year later than the person born in Barcelona.
- Daniel was born in Moscow.
- Anthony was born two years later than Edward was.

Which year was Judith born in?

A. 1975
B. 1976
C. 1977
D. 1978
E. Impossible to determine

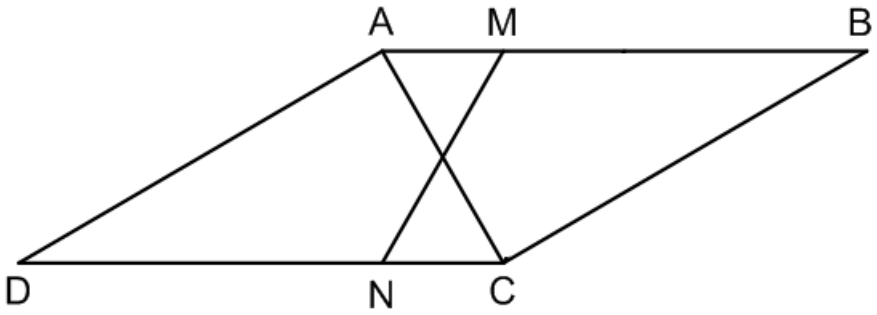
Question 14

A convex polyhedron has 48 edges. Its faces are either quadrilaterals or hexagons. Each vertex is an endpoint of exactly three edges. How many faces does the polyhedron have?

A. 16
B. 18
C. 20
D. 24
E. None of the above

Question 15

In the diagram below, $ABCD$ is a parallelogram and $\angle ACB = 90°$. Points M and N are on sides AB and CD such that $AM : MB = CN : ND = 1 : 3$. Given that $AC = MN = 2$, find the area of ABCD.



A. $2\sqrt{3}$
B. 6
C. $4\sqrt{3}$
D. 8
E. None of the above

Section B (Correct answer – 4 points| Incorrect or No answer – 0 points)

When an answer is a 1-digit number, shade "0" for the tens, hundreds and thousands place.
Example: if the answer is 7, then shade 0007
When an answer is a 2-digit number, shade "0" for the hundreds and thousands place.
Example: if the answer is 23, then shade 0023
When an answer is a 3-digit number, shade "0" for the thousands place.
Example: if the answer is 785, then shade 0785
When an answer is a 4-digit number, shade as it is.
Example: if the answer is 4196, then shade 4196

Question 16

Find the value of the following expression.

$$11^2 - 12^2 + 13^2 - 14^2 + 15^2 - 16^2 + \cdots + 99^2 - 100^2 + 101^2$$

Question 17

How many digits are there in $5^{2020} \times 4^{1010}$.

Question 18

The six-digit number 2X3Y72 is divisible by 66. Find the value of X + Y.

Question 19

Take a 2-digit number, multiply its digits, then multiply the digits of the obtained product and continue the process until you get a one-digit number. How many two-digit numbers will produce zero at the end of this process?

Question 20

Find the value of $(a+2b)(2b+3c)(a+3c)+6abc$ where $a+2b+3c=0$.

Question 21

Given that $P(x)=x^5-2021x^4+2021x^3-2021x^2+2021x-1$, what is the value of $P(2020)$?

Question 22

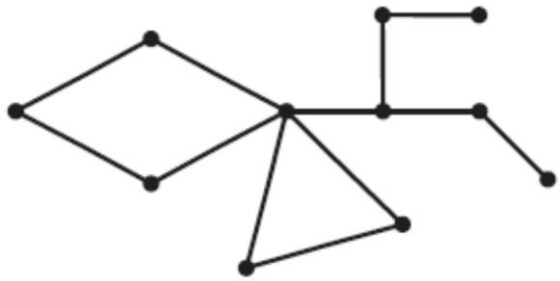
In the following cryptarithm, all the different letters stand for different digits.

$$\begin{array}{cccccccc} & C & I & R & C & L & E \\ & C & I & R & C & L & E \\ + & C & I & R & C & L & E \\ \hline & S & P & H & E & R & E \end{array}$$

Find the value of the sum S + P + H + E + R + E.

Question 23

In how many ways can the vertices in the diagram below be coloured with 3 colours such that any two neighbouring vertices are not coloured with the same colour?

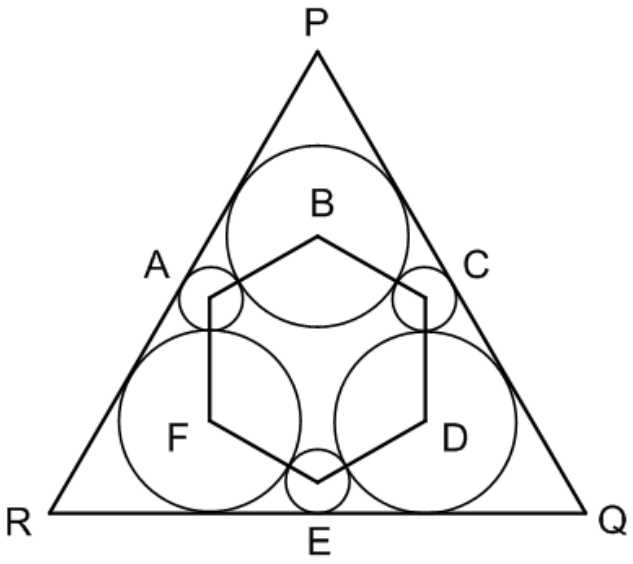


Question 24

Let $T_1 = 1$, $T_2 = 1 + \frac{1}{1}$, $T_3 = 1 + \frac{1}{1+\frac{1}{1}}$, ... , $T_n = 1 + \frac{1}{T_{n-1}}$. It is given that T_n can be expressed as a fraction in its simplest form, $T_n = \frac{P_n}{Q_n}$, where P_n and Q_n are positive coprime integers. How many numbers among P_1, P_2, P_3, ... , P_{2019} are odd?

Question 25

In the diagram below, there are 6 touching circles with centres in A, B, C, D, E and F. It is given that ABCDEF is a regular hexagon and AB = 4. PQR is a triangle such that each of its sides touches three of the circles. Find the square of the length of PQ.



END OF PAPER

Solutions to SASMO 2019 Secondary 3 (Grade 9)

Question 1

$$-\frac{1}{4}\times 2019.625 - 0.625 + 0.75 \times 2019.625 + \frac{1}{2}\times 2019.625$$
$$= 2019.625 \times \left(-\frac{1}{4} + 0.75 + \frac{1}{2}\right) - 0.625 = 2019.625 - 0.625 = \mathbf{2019}$$

Answer: **(A)**

Question 2

All the possible values of x are of the form $5^2 \times 11$ and $11^2 \times 5$. The smallest value of x is $5^2 \times 11 = \mathbf{275}$.

Answer: **(D)**

Question 3

Since $-11 \le y + 3 \le 3\ or\ -14 \le y \le 0$, then y is not positive. The greatest possible value of xy is **0** when $y = 0$.

Answer: **(C)**

Question 4

A. Wrong. $\sqrt{81} = 9$

B. Wrong. $(x^{12})^{18} = x^{12\times 18} = x^{216}$

C. Wrong. $5x^{-1} = \frac{5}{x}$

D. **True**. $-\frac{2(-x+1)(x+1)}{32} = -\frac{2(1-x)(1+x)}{32} = -\frac{1-x^2}{16} = \frac{1-x^2}{-2^4}$

Answer: **(D)**

Question 5

$$M \times N = S^1 \times W^2 \times A^3 \times P^4 \times P^1 \times A^2 \times W^3 \times S^4 = W^5 \times A^5 \times S^5 \times P^5$$
$$= (W \times A \times S \times P)^5 = \left((1331)^{\frac{1}{15}}\right)^5 = (1331)^{\frac{1}{3}} = \sqrt[3]{1331} = \mathbf{11}$$

Answer: **(A)**

Question 6

Option D statement is not true about a gradient since two lines are parallel if they have the same gradient.

Answer: **(D)**

Question 7

If the list $\{19, X, 13, 11, 23, 17\}$ has a mode, then $X = 11, 13, 17, 19\ or\ 23$.

When $X = 11$ and 13, the median is $\frac{13+17}{2} = 15$.

When $X = 19$ and 23, the median is $\frac{17+19}{2} = 18$.

When $X = 17$, the median is $\frac{17+17}{2} = 17$.

The greatest difference between the median and mode is $23 - 18 = \mathbf{5}$.

Answer: **(B)**

Question 8

$\frac{654321}{654322} = \frac{654322-1}{654322} = 1 - \frac{1}{654322}$

$\frac{65432}{65433} = \frac{65433-1}{65433} = 1 - \frac{1}{65433}$

$\frac{6543}{6544} = \frac{6544-1}{6544} = 1 - \frac{1}{6544}$

$\frac{654}{655} = \frac{655-1}{655} = 1 - \frac{1}{655}$

$\frac{65}{66} = \frac{66-1}{66} = 1 - \frac{1}{66}$

The largest fraction is $1 - \frac{1}{654322} = \mathbf{\frac{654321}{654322}}$.

Answer: **(A)**

Question 9

To get the least possible value of this integer, the first four digits should be 1, 0, 2 and 3 and the ones digit must be 8.

The 6-digit $1023A8$ is divisible by 9, then the sum of its digits $1 + 0 + 2 + 3 + A + 8 = 14 + A$ is divisible by 9. Hence $A = 4$ and the sum of its digits is **18**.

Answer: **(E)**

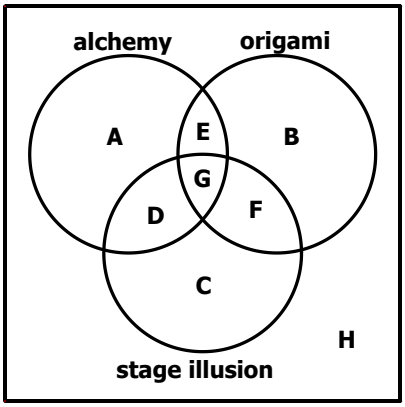
Question 10

$A = 10^{2019} - 2019 = \underbrace{99999999\ldots9}_{2015\ 9s}7981$ has $2015 + 3 =$ **2018** odd digits.

Answer: **(D)**

Question 11

Draw a Venn Diagram as shown in the diagram on the right.



A, B and C are the numbers of students who like only alchemy, origami and stage illusion, respectively. H is the number of students who have hobbies which are not listed.

13 like all the three hobbies or $\mathrm{G} = 13.$

45 have hobbies which are not listed or $\mathrm{H} = 45.$

24 like both alchemy and stage illusion or

$\mathrm{D} + \mathrm{G} = 24 \quad \Rightarrow \ \mathrm{D} = 24 - \mathrm{G} = 24 - 13 = 11.$

32 like both alchemy and origami or

$\mathrm{E} + \mathrm{G} = 32 \quad \Rightarrow \quad \mathrm{E} = 32 - \mathrm{G} = 32 - 13 = 19$

79 like alchemy or

$\mathrm{A} + \mathrm{E} + \mathrm{G} + \mathrm{D} = 79 \quad \Rightarrow \quad \mathrm{A} = 79 - \mathrm{E} - \mathrm{G} - \mathrm{D} = 79 - 19 - 13 - 11 = 36$

75 like origami or

$\mathrm{B}+\mathrm{F}+\mathrm{E}+\mathrm{G}=75 \quad \Rightarrow \quad \mathrm{B}+\mathrm{F}=75-\mathrm{E}-\mathrm{G}=75-19-13=43$

34 like stage illusion

$\mathrm{C}+\mathrm{F}+\mathrm{G}+\mathrm{D}=34 \quad \Rightarrow \quad \mathrm{C}+\mathrm{F}=34-13-11=10$

Gryffindor Secondary School has 170 students or

$$170=\mathrm{A}+\mathrm{B}+\mathrm{C}+\mathrm{D}+\mathrm{E}+\mathrm{F}+\mathrm{G}+\mathrm{H}=\mathrm{A}+(\mathrm{B}+\mathrm{F})+\mathrm{C}+\mathrm{D}+\mathrm{E}+\mathrm{G}+\mathrm{H}$$
$$=36+43+\mathrm{C}+11+19+13+45=167+\mathrm{C}$$

Hence $C=170-167=3$ and $\mathrm{F}=10-\mathrm{C}=7$

The number of Gryffindor students have hobbies which are both stage illusion and origami but not alchemy is $\mathrm{F}=$ **7**.

Answer: **(B)**

Question 12

If the 12th person is a truth-teller, then his/her number is bigger than 12. However, 12th person couldn't have responded, "My number is less than x", where $1 \leq x \leq 12$. Therefore, the 12th person is a liar. Also, two people who made the second responses

"My number is less than 1."

"My number is less than 2."

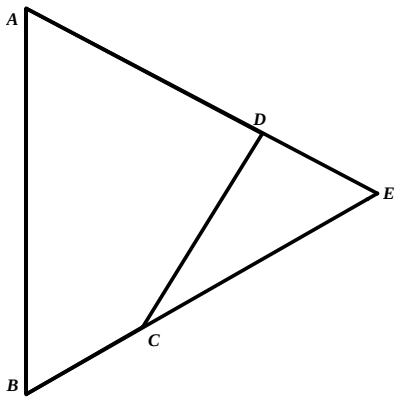
are liars as all the first responses describe the numbers bigger than 1.

The largest possible number of truth-tellers among these 12 people is **10**. It is possible when the 1st person got 2, the 2nd person got 3, …, the 11th person got 1 and the 12th person got 12.

Answer: **(B)**

Question 13

Let E be the intersection point of AD and BC.



$$\angle BEA = 180° - \angle EBA - \angle EAB = 180° - 60° - 60° = 60°$$

Hence the triangle ABE is equilateral and $AB = BE = EA$.

Triangles ACE and BDA are similar since $\angle CAE = \angle DBA$ and $\angle CEA = \angle DAB = 60°$. Therefore,

$$\frac{AE}{CE} = \frac{BA}{DA} \quad \Rightarrow \quad CE = DA = 13$$

$$AB = BE = BC + CE = 7 + 13 = \mathbf{20}.$$

Answer: **(A)**

Question 14

Assume that January 1, 2019 is a Monday. An ordinary year has $365 = (7 \times 52 + 1)$ days and a leap year has $366 = (7 \times 52 + 2)$ days. Thus, each year January 1st will shift back or forth by 1 or 2 days. When January 1st shifts by 7 or a multiple of 7 days in total, two calendars will be the same.

From 2019 to 2030, January 1st shifted forth by 14 days in total. Hence calendar years 2019 and **2030** are the same.

Year	2019	2020	2021	2022	2023	2024	2025	2026	2027
shift by		+1	+2	+1	+1	+1	+2	+1	+1

Year	2028	2029	2030	2031	2032	2033	2034	2035	2036
shift by	+1	+2	+1	+1	+1	+2			

Answer: **(B)**

Question 15

The figure doesn't have any square face. Therefore, the options B, C, D and E are not possible, and the answer is **option A**.

Answer: **(A)**

Question 16

Since $2018! = 2018 \times 2017 \times \ldots \times 2 \times 1$, it is clear that $2018!$ is divisible by $2019 = 673 \times 3$ (673 and 3 are relatively prime). Thus, when $2018! + 2018$ is divided by 2019, the remainder is $\mathbf{2018}$.

Answer: **2018**

Question 17

Since the product is an odd number, then the tens and ones digits must be odd numbers.

	Tens digit	Ones digit	Number of required 2-digit number
Number of options	5 $(1,3,5,7,9)$	5 $(1,3,5,7,9)$	$5 \times 5 = 25$

There are 5 options for tens digit. For each of these 5 options, there are 5 options for ones digits. Hence there are $5 \times 5 = \mathbf{25}$ two-digit numbers satisfying the given property.

Answer: **25**

Question 18

The product of 3 consecutive integers is $(a-1)a(a+1) = a(a^2-1) = a^3 - a$ which is close to perfect cube.

There are only 2 perfect cubes $16^3 = 4096$ and $17^3 = 4913$ between 4000 and 4999.

Hence there are only 2 possible triples (15, 16, 17) and (16, 17, 18). The last digit of $15 \times 16 \times 17$ is 0, and $16 \times 17 \times 18 = 4896$ which satisfies the question statement. The sum of these 3 consecutive integers is $16 + 17 + 18 = \mathbf{51}$.

Answer: **51**

Question 19

$\overline{bbbbb} = 11111b$ and $\overline{bb} = 11b$

Rewrite the expression as

$$\frac{(b+b)\times(11111b-b)-b\times 11b}{b\times 11b} = \frac{2b\times 11110b-11b^2}{11b^2} = \frac{22220b^2-11b^2}{11b^2}$$
$$= \frac{22220b^2-11b^2}{11b^2} = \frac{22209b^2}{11b^2} = \mathbf{2019}$$

Answer: **2019**

Question 20

$$\frac{1-\left(\frac{1}{a}\right)^2}{1+\left(\frac{1}{a}\right)^2} = \frac{\frac{a^2-1}{a^2}}{\frac{a^2+1}{a^2}} = \frac{a^2-1}{a^2+1}$$

$$\frac{1-\left(\frac{1}{a}\right)^2}{1+\left(\frac{1}{a}\right)^2} + \frac{1-a^2}{a^2+1} = \frac{a^2-1}{a^2+1} + \frac{1-a^2}{a^2+1} = 0$$

$$\frac{1-\left(\frac{1}{2019}\right)^2}{1+\left(\frac{1}{2019}\right)^2} + \frac{1-\left(\frac{1}{2018}\right)^2}{1+\left(\frac{1}{2018}\right)^2} + \cdots + \frac{1-\left(\frac{1}{2}\right)^2}{1+\left(\frac{1}{2}\right)^2} + \frac{1-1^2}{1+1^2} + \frac{1-2^2}{1+2^2} + \cdots + \frac{1-2019^2}{1+2019^2}$$
$$= \left(\frac{1-\left(\frac{1}{2019}\right)^2}{1+\left(\frac{1}{2019}\right)^2} + \frac{1-2019^2}{1+2019^2}\right) + \left(\frac{1-\left(\frac{1}{2018}\right)^2}{1+\left(\frac{1}{2018}\right)^2} + \frac{1-2018^2}{1+2018^2}\right) + \cdots$$
$$+ \left(\frac{1-\left(\frac{1}{2}\right)^2}{1+\left(\frac{1}{2}\right)^2} + \frac{1-2^2}{1+2^2}\right) + \frac{1-1^2}{1+1^2} = \mathbf{0}$$

Answer: **0**

Question 21

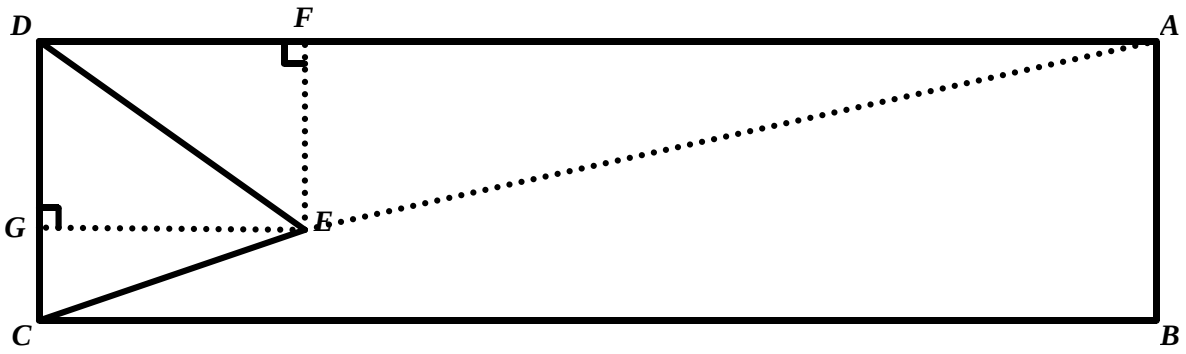
The prime factorisation of 504 is $2^3 \times 3^2 \times 7$. If X is a 6-digit number, then the smallest product of its digit must be $1 \times 2 \times 3 \times 4 \times 5 \times 6 = 720$. Hence X must be a 5-digit number so that the value of $\frac{X}{10}$ is the greatest possible.

The greatest possible value of X is 97421 and the product of its digits is $9 \times 7 \times 4 \times 2 \times 1 = 504$. Rounding off $\frac{X}{10}$ to the nearest integer, we get **9742**.

Answer: **9742**

Question 22

Let EF and EG be the perpendicular lines to AD and CD, respectively, as shown below.



Using Cosine Rule in triangle DEC,

$EC^2 = DC^2 + DE^2 - 2DC \times DE \times \cos\angle CDE$ or

$25^2 = 25^2 + 30^2 - 2 \times 25 \times 30 \times \cos\angle CDE \quad \Rightarrow$

$2 \times 25 \times 30 \times \cos\angle CDE = 30^2$ or

$\cos\angle CDE = \frac{30^2}{2 \times 25 \times 30} = \frac{3}{5}$

In triangle DGE,

$$\cos\angle CDE = \frac{DG}{DE} = \frac{DG}{30} = \frac{3}{5}$$

Hence $DG = 18 = EF$.

Using Pythagoras Theorem,

$$DF^2 = DE^2 - EF^2 = 30^2 - 18^2 = 576 = 24^2 \quad \Rightarrow \quad DF = 24.$$

$$AF = AD - DF = 104 - 24 = 80$$

Using Pythagoras Theorem again,

$$AE^2 = AF^2 + EF^2 = 80^2 + 18^2 = 6724 = 82^2 \quad \Rightarrow \quad AE = \mathbf{82}$$

Answer: **82**

Question 23

The number 9 cannot be between 2 numbers as there is only 1 way to get perfect square which is $9 + 7 = 16$. So 9 must be the first or the last number in a row.

When 9 is the first number, there is only one arrangement:

$$9, 7, 2, 14, 11, 5, 4, 12, 13, 3, 6, 10, 15, 1, 8$$

When 9 is the last number, there is also only one arrangement:

$$8, 1, 15, 10, 6, 3, 13, 12, 4, 5, 11, 14, 2, 7, 9$$

There are **2** arrangements.

Answer: **2**

Question 24

Expand the equation

$$x^3 + y^3 + x^2y^2 + xy = x^3 + 3x^2y + 3xy^2 + y^3 \quad \Rightarrow \quad x^2y^2 + xy = 3x^2y + 3xy^2$$

Next, divide both side by xy

$$xy + 1 = 3x + 3y$$

$$xy - 3x = 3y - 1$$

$$x(y - 3) = 3y - 1$$

$$x = \frac{3y - 1}{y - 3} \quad \Rightarrow \quad y > 3,$$

for x to be a positive integer

Without loss of generality, assume that $x \geq y$. Then

$$xy + 1 = 3x + 3y \leq 6x$$

$$6x - xy \quad \geq 1$$

$$x(6 - y) \geq 1$$

Since $x > 0$, then $6 - y > 0$. This implies $y < 6$ or $y \leq 5$.

Thus, there are 2 solutions $x = 7, y = 5$ and $x = 11, y = 4$. The greatest possible value of $x + y$ is $11 + 4 = \mathbf{15}$.

Answer: **15**

Question 25

It is clear that S = 1. To find the value of S + A + S + M + O, we need to find the values of A and O.

In the tens place addition,

$$O + T = 17, 16, 7 \text{ or } 6$$

Case 1: $O + T = 16$ doesn't have any solutions. Case 1 is impossible.

Case 2: If $O + T = 6$, then $O + A = 11$. $O + T = 6$ has 2 possible solutions $O = 2 \; or \; 4$. Since $A \neq 7$, then $O \neq 4$. Hence $O = 2 \;\; \Rightarrow \;\; A = 9$ which is impossible in thousands place addition. Case 2 is impossible.

Case 3: $O + T = 7$, then $O + A = 11$. $O + T = 7$ has 4 possible solutions:

T	O	A	C	**Remarks**
5	2	9	11	Impossible. C must be digit
4	3	8	10	Impossible. C must be digit
3	4	7	9	Impossible $A \neq M$
2	5	6	8	might be possible

If $O + T = 7$, then $L + H < 10$ and $L + H = O = 5$. However, there is no solution when $L + H = 5$ since L and H cannot be 1 or 2. Case 3 is impossible.

Case 4: if $O + T = 17$, then $O + A = 10$. $O + T = 17$ has 2 possible solutions:

T	O	A	C	
8	9	1		Impossible. $A \neq S$
9	8	2	4	Possible.

The value of S + A + S + M + O is $1 + 2 + 1 + 7 + 8 = \mathbf{19}$.

Answer: **19**

Solutions to SASMO 2020 Secondary 3 (Grade 9)

Question 1

$$2020 \times 2020 - 2020 \times 20 + 2020 = 2020 \times (2020 - 20 + 1) = 2020 \times 2001$$
$$= \mathbf{4042020}$$

Answer: **(B)**

Question 2

A. 2020

B. $\frac{2020^{20}}{2020^{18}} = 2020^2 = 4080400$

C. 4080400

D. $\frac{2020^{20}-2020}{2020^{18}-2020} = \frac{2020^{19}-1}{2020^{17}-1} = \frac{2020^{19}-2020^2+2020^2-1}{2020^{17}-1} = \frac{2020^{19}-2020^2}{2020^{17}-1} + \frac{2020^2-1}{2020^{17}-1}$
$$= \frac{2020^2(2020^{17}-1)}{2020^{17}-1} + \frac{2020^2-1}{2020^{17}-1} = 2020^2 + \frac{2020^2-1}{2020^{17}-1}$$

Option D has the largest value.

Answer: **(D)**

Question 3

$11! - 10! = 10!\,(11-1) = 10 \times 10!$

$10!$ contains 2 pairs of (2,5) in its product , hence 10! has two consecutive ending zeros and $10 \times 10!$ has **3** consecutive zeros**.**

Answer: **(E)**

Question 4

$$\frac{2025 \times (19!)^2 - 5 \times (19!)^2}{100 \times 18! \times 20! + 18! \times 20!} = \frac{(19!)^2 \times (2025-5)}{18! \times 20! \times (100+1)} = \frac{(18! \times 19) \times 19! \times 2020}{18! \times (19! \times 20) \times 101}$$
$$= \frac{19 \times 2020}{20 \times 101} = \mathbf{19}$$

Answer: **(A)**

Question 5

$$91^4 - 1 = (91^2 - 1)(91^2 + 1) = 8280 \times 8282 = 20 \times 414 \times 101 \times 82$$
$$= 20 \times 101 \times 414 \times 82 = \mathbf{2020} \times 414 \times 82$$

Answer: **(B)**

Question 6

1 cannot be one of the six positive integers as 1 is coprime to all positive integers. Then each of the six positive integers must be greater than 1. Thus, the smallest positive desired integer is $\mathbf{64} = 2 \times 2 \times 2 \times 2 \times 2 \times 2$.

Answer: **(B)**

Question 7

As each pair of friends requires a separate social network, the result cannot be less than $\binom{4}{2} = 6$. It's possible to construct a solution with exactly 6 social networks. For example, if we call the friends as A, B, C, D and number the social networks by numbers from 1 to **6**, then the solution can be {A: (1, 2, 3), B: (1, 4, 5), C: (2, 4, 6), D: (3, 5, 6)}.

Answer: **(C)**

Question 8

It is given that

$$f(-1) = a - b + c = 0 \quad (1)$$

$$11 = f(2) - f(1) = (4a + 2b + c) - (a + b + c) = 3a + b \quad (2)$$

$$15 = f(3) - f(2) = (9a + 3b + c) - (4a + 2b + c) = 5a + b \quad (3)$$

Solving equations (2) and (3), we get $a = 2$ and $b = 5$. From equation (1), $c = b - a = 5 - 2 = 3$. The value of $a + b + c$ is $2 + 5 + 3 = \mathbf{10}$.

Answer: **(E)**

Question 9

Each number is the sum of the squares of digits of the previous number:

$37 = 1^2 + 6^2$

$58 = 3^2 + 7^2$

$89 = 5^2 + 8^2$

$145 = 8^2 + 9^2$

$42 = 1^2 + 4^2 + 5^2$

The next number in the sequence is $4^2 + 2^2 = \mathbf{20}$.

Answer: **(A)**

Question 10

This is an example of a well-known balancing puzzle. It can be proved that for n coins the solution is $\lceil \log_3 n \rceil$, using a divide-and-conquer strategy: put one third of the coins on one pan and the same number of coins on the other pan. Therefore, the answer is **7**.

Answer: **(A)**

Question 11

The total number of 4-digit numbers is 9000:

Thousands place	Hundreds place	Tens place	Ones place	Total
9 options (1, 2, 3, 4, …, 9)	10 options (0, 1, 2, 3, 4, …, 9)	10 options (0, 1, 2, 3, 4, …, 9)	10 options (0, 1, 2, 3, 4, …, 9)	$9 \times 10 \times 10 \times 10 = 9000$

The number of 4-digit numbers with no digit '2's is 5832:

Thousands place	Hundreds place	Tens place	Ones place	Total
8 options (1, 3, 4, …, 9)	9 options (0, 1, 3, 4, …, 9)	9 options (0, 1, 3, 4, …, 9)	9 options (0, 1, 3, 4, …, 9)	$8 \times 9 \times 9 \times 9 = 5832$

Hence, the number of 4-digit numbers which contain digit '2' is $9000 - 5832 = 3168$.

The number of 4-digit numbers which do not contain digit '1' and '2' is 3584:

Thousands place	Hundreds place	Tens place	Ones place	Total
7 options (3, 4, …, 9)	8 options (0, 3, 4, …, 9)	8 options (0, 3, 4, …, 9)	8 options (0, 3, 4, …, 9)	$7 \times 8 \times 8 \times 8 = 3584$

Hence, the number of 4-digit numbers which contain digit '1' or '2' is $9000 - 3584 = 5416$.

Thus, the number of four-digit numbers which have digit 1 but not digit 2 among its digits is $5416 - 3168 = \mathbf{2248}$.

Answer: **(A)**

Question 12

$$\angle ADE = 90° - \angle EAD = 90° - \angle CAB = \angle BCA = \angle GCF$$

$$\angle EAD = 90° - \angle ADE = 90° - \angle GCF = \angle FGC$$

Thus, triangles AED and GFC are similar and

$$\frac{AE}{DE} = \frac{GF}{FC} \Leftrightarrow \frac{4}{6} = \frac{6}{FC} \Leftrightarrow FC = 9.$$

Quadrilateral $ACGD$ is trapezium and its area is

$$DE \times \frac{(GD + AC)}{2} = 6 \times \frac{(6 + 19)}{2} = \mathbf{75}.$$

Answer: **(B)**

Question 13

From the first two statements, we get that the person born in Barcelona is one year older than Edward who was born in Singapore.

From the 4th statement, Anthony is two years younger than Edward (from Singapore) and three years younger than the person born in Barcelona. Hence Anthony was born in 1978 in New York since Daniel was born in Moscow.

Judith was born in **1975** in Barcelona.

Answer: **(A)**

Question 14

As every vertex belongs to three edges, and every edge connects two vertices, the total number of vertices is equal to $48 \times 2 \div 3 = 32$.

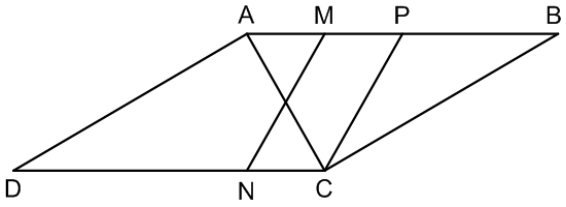
Since the polyhedron is convex and therefore simply connected, we can use Euler's formula. Using Euler's Formula, $F + V = E + 2$, where F is the number of faces, V is the number of vertices, and E is the number of edges, we get

$$F = E - V + 2 = 48 - 32 + 2 = \mathbf{18}.$$

Answer: **(B)**

Question 15

Let us draw segment CP parallel to MN where P lies on AB. Since $MP = NC = AM = \frac{1}{4}AB$, then $AP = PB = \frac{1}{2}AB$. Since $CP = MN = AC$ and $\angle ACB$ is a right angle, then triangle ABC is a right-angled triangle where the length of one of its legs is equal to the half the length of its hypotenuse. Thus, triangle ABC has angles 60, 30 and 90 degrees. Therefore, the length of BC is $\sqrt{3} \times AC = 2\sqrt{3}$, and the area of ABCD is $2 \times \frac{1}{2} \times 2 \times 2\sqrt{3} = \mathbf{4\sqrt{3}}$.



Answer: **(C)**

Question 16

$$11^2 - 12^2 + 13^2 - 14^2 + 15^2 - 16^2 + \cdots + 99^2 - 100^2 + 101^2 =$$

$$= (11 - 12) \times (11 + 12) + (13 - 14) \times (13 + 14) + \cdots + (99 - 100) \times (99 + 100) + 101^2 = -1 \times (11 + 12 + \cdots + 99 + 100) + 101^2 =$$

$$= -1 \times \left(\frac{100 \times 101}{2} - \frac{10 \times 11}{2}\right) + 101^2 = 101^2 - 50 \times 101 - \left(-(5 \times 11)\right) =$$

$$= 101 \times (101 - 50) + 55 = \mathbf{5206}$$

Answer: **5206**

Question 17

$$5^{2020} \times 4^{1010} = 5^{2020} \times (2^2)^{1010} = 5^{2020} \times 2^{2020} = (5 \times 2)^{2020} = 10^{2020}$$

10^{2020} has **2021** digits.

Answer: **2021**

Question 18

The six-digit number 2X3Y72 must be divisible by 2, 3 and 11.

From the Divisibility Rule of 11:

$|(2+3+7)-(X+Y+2)| = |10-X-Y| = 0$ or $X+Y = \mathbf{10}$.

Answer: **10**

Question 19

First, we list down all 2-digit numbers that produce zero after one operation.

After 1 operation: 10, 20, 30, 40, 50, 60, 70, 80, 90 – 9 numbers

Next, we list down the 2-digit numbers that will produce the 2-digit numbers listed above.

25, 52, 45, 54, 56, 65, 58, 85 – 8 numbers

Finally, we list down the 2-digit numbers which will produce the 2-digit number in the previous list of 8 numbers.

55, 59, 95, 96, 69, 87, 78 – 7 numbers.

The process is stopped here as no two-digit numbers can produce any of the 7 numbers in the last list.

Thus, there are $9+8+7 = \mathbf{24}$ such two-digit numbers.

Answer: **24**

Question 20

From the equation $a+2b+3c=0$, we get $(a+2b) = -3c, (2b+3c) = -a$ and $(a+3c) = -2b$. Thus,

$$(a+2b)(2b+3c)(a+3c)+6abc = (-3c)\times(-a)\times(-2b)+6abc = -6abc+6abc$$
$$= \mathbf{0}$$

Answer: **0**

Question 21

$$\begin{aligned} P(x) &= x^5 - 2021x^4 + 2021x^3 - 2021x^2 + 2021x - 1 \\ &= x^5 - 2020x^4 - x^4 + 2020x^3 + x^3 - 2020x^2 - x^2 + 2020x + x - 1 \\ &= x^4(x - 2020) - x^3(x - 2020) + x^2(x - 2020) - x(x - 2020) + x - 1 \end{aligned}$$

Thus,

$$P(2020) = 2020 - 1 = \mathbf{2019}$$

Answer: **2019**

Question 22

Rewrite the sum as the product:

	C	I	R	C	L	E
×						3
	S	P	H	E	R	E

In ones place, the letter E must be 5. (E=0 can be checked similarly, and it doesn't lead to any solution.)

The six-digit number CIRCLE times 3 results in a 6-digit number, therefore C=1, 2 or 3.

If C=3, then in hundreds place product $C \times 3$, there must be a carry over of 6 from tens place product which is impossible.

If C=2, then in hundreds place product $C \times 3$, there must be a carry over of 9 from tens place product which is impossible. Hence C=1.

In hundreds place product $C \times 3$, there must be a carry over of 2 from tens place product which implies that L=8 or 9.

If L=8, then R=5=E which is impossible. Hence L=9, R=8 and H=4.

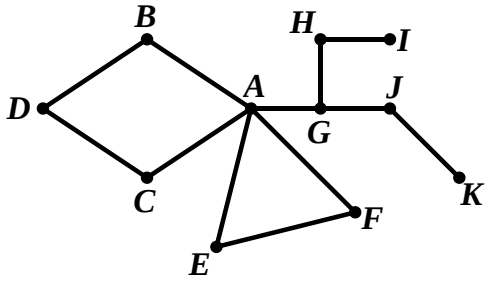
In hundred thousands place product, S cannot be 4 and 5 since H=4 and E=5. Hence S=3 and the only one-digit number left for I is 0.

The value of the sum S + P + H + E + R + E is $3 + 2 + 4 + 5 + 8 + 5 = \mathbf{27}$.

Answer: **27**

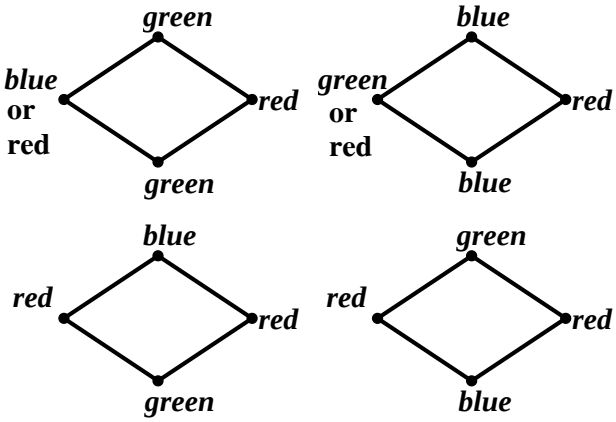
Question 23

Label the vertices as shown below.

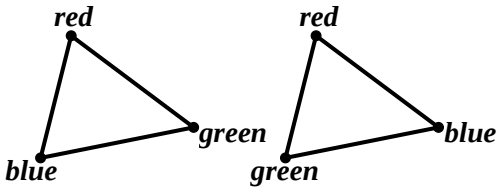


Let us colour point A with red colour.

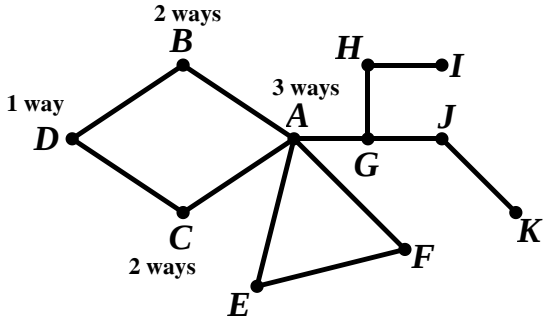
There are 6 ways to colour quadrilateral ABDC.



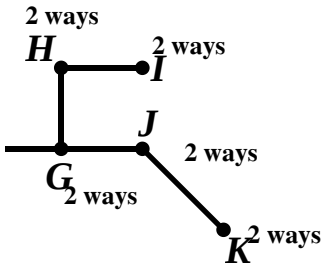
There are 2 ways to colour triangle GEF.



Point A – 3 ways, points B and C – 2 ways each since one colour was used on point A and point D – 1 way since two colours were used on points B and C.



There are $2^5 = 32$ ways to colour the GHIJK.



Thus, when point A is red, there are $6 \times 2 \times 32 = 384$ ways. Similarly, there are 384 ways each to colour the entire figure when point A is blue or green.

Hence, the total number of ways to colour the figure is $384 \times 3 = \mathbf{1152}$.

Answer: **1152**

Question 24

First, calculate some T_n values when n is small:

$$T_1 = \frac{1}{1}, T_2 = \frac{2}{1}, T_3 = \frac{3}{2}, T_4 = \frac{5}{3}, T_5 = \frac{8}{5}.$$

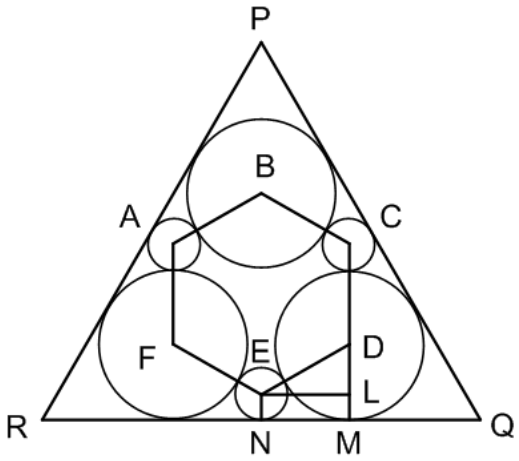
Note that both numerators and denominators correspond to Fibonacci numbers. Using mathematical induction, it can easily be proved that $P_n = P_{n-1} + P_{n-2}$ and $Q_n = P_{n-1}$ for every $n \geq 3$.

We can notice that $P_1 = odd$, $P_2 = even$, $P_3 = odd$, $P_4 = odd$, $P_5 = even$, $P_6 = odd$ and so on where the sequence $odd, even, odd$ will keep repeating. Thus, there are $2019 \div 3 \times 2 = \mathbf{1346}$ odd numbers among P_1, P_2, P_3, ..., P_{2019}.

Answer: **1346**

Question 25

Draw lines DM and EN perpendicular to Q, where M and N lie on QR, and line EL parallel to QR as shown.



Since $ABCDEF$ is a regular hexagon, then PQR is also a regular triangle and CD is perpendicular to QR. Therefore C, D and M lie on a straight line, $\angle EDL = 180° - \angle CDE = 60°$ and $DL = \frac{ED}{2}$.

Since $EN + DM = ED$, we have

$$DM = ED - EN \Rightarrow$$

$DL = DM - LM = (ED - EN) - EN = ED - 2EN \Rightarrow \frac{ED}{2} = ED - 2EN \Rightarrow EN = \frac{ED}{4} = \frac{AB}{4} = 1$

Also, $NM = EL = \frac{\sqrt{3}}{2} \times ED = 2\sqrt{3}$ and $MQ = \sqrt{3}DM = 3\sqrt{3}$. Therefore, $QR = PQ == 2 \times (2\sqrt{3} + 3\sqrt{3}) = 10\sqrt{3}$, and the square of its length is $(10\sqrt{3})^2 = \mathbf{300}$.

Answer: **300**